

Local Orbit Consistency for Symmetry-Robust Neural Operators

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Abstract

Neural operators can learn accurate PDE solution maps while still violating known symmetries of the underlying equations. We study a model-agnostic remedy: stochastic local orbit consistency, a training objective that compares the prediction on a transformed input with the transformed original prediction. The method uses known Lie-group actions, requires no architectural change, supports labeled training and unlabeled adaptation, and leaves inference identical to the base operator. In a periodic N64 2D Navier–Stokes Galilean setting, augmentation plus orbit consistency reduces symmetry-induced OOD error and equivariance defect relative to supervised and augmentation baselines, with the strongest gains in low-label regimes.

1 Introduction

Neural operators are attractive for PDE learning because they approximate maps between function spaces rather than single finite-dimensional predictions. This flexibility also exposes a weakness: a high-accuracy operator need not respect symmetries that the exact solution operator satisfies. When test inputs move along a known symmetry orbit, such as a Galilean boost of a periodic flow, ordinary supervised training can produce both higher prediction error and larger equivariance defect.

We propose to reduce this failure mode with local orbit consistency. Given a known group action on inputs and outputs, the objective samples a small finite transform and penalizes the discrepancy between $\mathcal{G}_\theta(T_g a)$ and $T_g \mathcal{G}_\theta(a)$. The penalty is a training-time regularizer, not a new inference pipeline. This makes it complementary to supervised Lie augmentation and architecture-level equivariance: it can be added to an existing FNO, applied in low-label settings, and evaluated on unlabeled inputs.

Our empirical claim is deliberately narrow. On exact periodic 1D and 2D symmetry tests, the coded group actions satisfy solver-level identities. On the main N64 2D Galilean experiment, augmentation plus orbit consistency with the same FNO2d backbone lowers orbit OOD relative L^2 and equivariance defect compared with compute-matched supervised augmentation. The final artifact path, manifest, tables, and figures are regenerated by `python scripts/reproduce_paper_artifacts.py`.

2 Related Work

Neural operators learn mappings between function spaces and are therefore a natural model class for parametric PDE solution operators. DeepONet introduced a branch–trunk architecture for learning nonlinear operators from function samples and output coordinates [13]. Fourier Neural Operators (FNOs) parameterize global integral kernels in Fourier space and provide an efficient backbone for PDE families on regular grids [9]. The general neural-operator framework formalizes these models as discretization-invariant compositions of integral operators and nonlinearities, with applications to Burgers’ equation, Darcy flow, and Navier–Stokes [7]. Physics-Informed Neural Operators (PINOs) add PDE residual constraints to operator learning, combining supervised operator regression with physics-based optimization [12]. Subsequent work extends neural operators to geometric settings: Geo-FNO learns deformations from irregular physical domains to latent uniform grids [10], GINO combines graph and Fourier operator components for large-scale three-dimensional PDEs on varying geometries [11], and NORM uses Laplacian eigenfunctions to define neural operators on Riemannian manifolds [3]. Operator-learning benchmarks such as PDEBench and The Well provide reusable datasets and baselines for time-dependent PDEs and broader physical simulation families [14, 17]. This literature establishes FNO-style models as computationally efficient solution-operator backbones, while also showing that domain geometry, discontinuities, and discretization effects require additional structure [8].

Symmetry has been used as an inductive bias through architectures, losses, and data transformations. Group-equivariant convolutional networks replace ordinary convolution by group convolution to share weights across symmetry orbits [4]; steerable CNNs generalize this principle through representation-constrained kernels [19]; and gauge-equivariant CNNs extend equivariance from global transformations to local frame changes on manifolds [5]. For operator learning, Group-Equivariant FNOs encode translation, rotation, and reflection symmetries in Fourier layers [6]. A parallel line derives symmetries from the governing differential equations. Lie Point Symmetry Data Augmentation constructs solution-preserving transformations from a PDE’s Lie point symmetry group and uses them to reduce sample complexity for neural PDE solvers [2]. Lie-symmetry losses for PINNs enforce conservation of solution families under infinitesimal generators [1]. Generalized Lie symmetries for PINOs further argue that evolutionary representatives can provide stronger training signals than point symmetries alone [18]. These works support the central premise that PDE symmetries should constrain the learned solution operator, not only the training data distribution.

Model-agnostic symmetry methods provide the closest precedent for the proposed direction. Tangent propagation used transformation tangents to impose local invariance constraints on neural networks [16]. Lie Algebra Canonicalization uses infinitesimal generators to construct canonicalized inputs for pretrained models and neural operators under arbitrary Lie groups, including non-compact groups commonly arising in PDEs [15]. PACE-FNO applies a related canonicalization strategy to FNOs under symmetry-induced shifts, estimating a Lie-algebraic input frame, predicting in a canonical frame, and restoring the output frame for translations, rotations, and Galilean boosts [20]. Our method studies a complementary point in this design space: a stochastic local orbit-consistency objective that acts directly on an existing neural operator. The regularizer uses known infinitesimal generators, evaluates finite group perturbations, supports labeled training and unlabeled adaptation, and leaves the inference map identical to the base operator.

3 Method

We train a neural operator $\mathcal{G}_\theta : \mathcal{A} \rightarrow \mathcal{U}$ with ordinary supervised regression and a stochastic local orbit-consistency objective. Each minibatch contains input–solution pairs (a, u) . The supervised term is

$$\mathcal{L}_{\text{sup}}(\theta) = \mathbb{E}_{(a,u)} \left[\frac{\|\mathcal{G}_\theta(a) - u\|_{\mathcal{U}}^2}{\|u\|_{\mathcal{U}}^2 + \delta} \right]. \quad (1)$$

For a sampled Lie-algebra generator X_i , scalar step ϵ , and finite transform $g = \exp(\epsilon X_i)$, the local orbit-consistency term is

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a,i,\epsilon} \left[\frac{\|\mathcal{G}_\theta(T_g^{\mathcal{A}} a) - T_g^{\mathcal{U}} \mathcal{G}_\theta(a)\|_{\mathcal{U}}^2}{\epsilon^2 + \eta} \right]. \quad (2)$$

The full objective is $\min_\theta \mathcal{L}_{\text{sup}}(\theta) + \lambda \mathcal{L}_{\text{orb}}(\theta)$. For supervised Lie augmentation, we also evaluate $\mathcal{L}_{\text{aug}}(\theta) = \mathbb{E}_{(a,u),g} [\|\mathcal{G}_\theta(T_g^{\mathcal{A}} a) - T_g^{\mathcal{U}} u\|_{\mathcal{U}}^2 / (\|T_g^{\mathcal{U}} u\|_{\mathcal{U}}^2 + \delta)]$, and report both augmentation-only and augmentation-plus-orbit variants.

The method is architecture-agnostic. In our experiments, $\mathcal{G}_\theta$ is an FNO backbone. Transformations are applied to physical-grid input fields; the prediction on the transformed input is compared with the transformed original prediction; gradients pass through both model evaluations. Integer periodic translations are tensor rolls, non-integer periodic translations are spectral shifts, and finite square symmetries are exact tensor rotations and flips. For 1D Burgers, the Galilean action on the solution map $u_0 \mapsto u_T$ is $T_c^{\mathcal{A}} u_0(x) = u_0(x) + c$ and $T_c^{\mathcal{U}} u_T(x) = u_T(x - cT) + c$.

The same loss supports unlabeled adaptation. Given a pretrained operator $\mathcal{G}_{\theta_0}$ and target inputs $a \sim \mathcal{D}_{\text{target}}$, we minimize $\mathcal{L}_{\text{orb}}^{\text{target}}(\theta) + \beta \|\theta - \theta_0\|_2^2$ over a chosen adaptation set Θ_{adapt} , such as all parameters, the last operator block, or the projection head. This adaptation objective requires no PDE solves and no target labels.

4 Theory

Let $\mathcal{A}$ and $\mathcal{U}$ denote Banach spaces of input and output fields. A PDE family induces a solution operator $\mathcal{S} : \mathcal{A} \rightarrow \mathcal{U}$. Let G be a Lie group acting on the full problem family through transformations $T_g^{\mathcal{A}}$ and $T_g^{\mathcal{U}}$.

Closure of the full problem class under G means that the equation, coefficients, forcing terms, boundary or initial conditions, domain class, and observation map transform consistently. Under this assumption the exact solution operator is equivariant: $\mathcal{S}(T_g^A a) = T_g^U \mathcal{S}(a)$. For a learned operator $\mathcal{G}_\theta$, define the operator-level equivariance defect $\Delta_\theta(a, g) = \mathcal{G}_\theta(T_g^A a) - T_g^U \mathcal{G}_\theta(a)$. Our regularizer samples $g = \exp(\epsilon X_i)$ and penalizes

$$\mathcal{L}_{\text{orb}}(\theta) = \mathbb{E}_{a, i, \epsilon} \left[\frac{\|\Delta_\theta(a, \exp(\epsilon X_i))\|_{\mathcal{U}}^2}{\epsilon^2 + \eta} \right]. \quad (3)$$

The objective is label-free because both terms are model predictions transformed by known group actions.

Infinitesimal interpretation. Assume $T_{\exp(\epsilon X)}^A$, $T_{\exp(\epsilon X)}^U$, and $\mathcal{G}_\theta$ are differentiable at $\epsilon = 0$. Let $X^A a$ and $X^U u$ denote the induced infinitesimal actions. Taylor expansion gives

$$\mathcal{G}_\theta(T_{\exp(\epsilon X)}^A a) - T_{\exp(\epsilon X)}^U \mathcal{G}_\theta(a) = \epsilon (D\mathcal{G}_\theta(a)[X^A a] - X^U \mathcal{G}_\theta(a)) + O(\epsilon^2). \quad (4)$$

Thus $\|\Delta_\theta(a, \exp(\epsilon X))\|_{\mathcal{U}}^2 / \epsilon^2$ converges to $\|D\mathcal{G}_\theta(a)[X^A a] - X^U \mathcal{G}_\theta(a)\|_{\mathcal{U}}^2$ as $\epsilon \rightarrow 0$. The finite-orbit objective is therefore a stochastic finite-difference estimator of the squared Lie-algebraic equivariance defect, while avoiding explicit Jacobian-vector products.

Orbit error decomposition. If the true solution operator is equivariant and T_g^U is bounded, then for any $a \in \mathcal{A}$ and $g \in G$,

$$\begin{aligned} \|\mathcal{G}_\theta(T_g^A a) - \mathcal{S}(T_g^A a)\|_{\mathcal{U}} &\leq \|\mathcal{G}_\theta(T_g^A a) - T_g^U \mathcal{G}_\theta(a)\|_{\mathcal{U}} \\ &\quad + \|T_g^U\| \|\mathcal{G}_\theta(a) - \mathcal{S}(a)\|_{\mathcal{U}}. \end{aligned} \quad (5)$$

The first term is the learned equivariance defect and the second term is the in-distribution prediction error transported by the output action. This inequality motivates reducing $\mathcal{L}_{\text{orb}}$ when test inputs differ from training inputs by known PDE symmetries.

Approximate symmetries and cost. For partial or approximate symmetries, replace the output norm by a weighted norm $\|v\|_w^2 = \|w_{a,g} \odot v\|^2$, where $w_{a,g}$ can encode valid overlap after transformations, boundary exclusions, observation masks, or regions where transformed coefficients and forcing terms remain physically meaningful. With one sampled transform per minibatch, training uses one ordinary forward pass, one transformed forward pass, and a known output transformation. Inference is exactly the base map $\mathcal{G}_\theta(a)$, so deployment latency and memory are unchanged relative to the underlying neural operator.

5 Experiments

Question. We test whether local orbit consistency reduces symmetry-induced out-of-distribution error and equivariance defect for an unchanged neural-operator backbone. The main experiment uses a periodic 64×64 2D Navier–Stokes vorticity operator with Galilean boosts. The base model is the same FNO2d architecture in every row; orbit consistency changes training only and does not add an inference-time module.

Protocol. We compare four methods: supervised FNO, supervised Lie augmentation, supervised orbit consistency, and augmentation plus orbit consistency. The headline low-label setting uses 2% labeled data and five seeds. The compute-matched augmentation baseline uses six update steps per epoch, while augmentation plus orbit consistency uses four update steps per epoch with $\lambda_{\text{orb}} = 0.10$. All reported rows use the same test-time model architecture and are regenerated by `python scripts/reproduce_paper_artifacts.py`.

Main N64 Galilean result. Table 1 shows that at 2% labels, augmentation plus orbit consistency improves both paper-facing symmetry metrics over compute-matched supervised augmentation. Orbit OOD relative L^2 decreases from 0.5276 ± 0.0525 to 0.3950 ± 0.0334 , a 25.1% reduction. Equivariance defect decreases from 0.3816 ± 0.0322 to 0.2677 ± 0.0174 , a 29.8% reduction. Relative to the plain supervised FNO baseline, the reductions are 58.7% for orbit OOD error and 61.7% for equivariance defect. Parameter counts are identical across the headline rows.

Table 1: N64 Galilean headline at 2% labels over five seeds. All rows use the same FNO2d architecture with 2,668,609 parameters. Lower is better for relative L^2 , orbit OOD relative L^2 , and equivariance defect.

Method	Steps/epoch	Relative L^2	Orbit OOD L^2	Eq. defect	Latency ms/sample
FNO	4	0.9287 ± 0.0228	0.9568 ± 0.0223	0.6988 ± 0.0173	16.53 ± 2.13
FNO + orbit	4	0.4646 ± 0.0394	0.4952 ± 0.0348	0.3359 ± 0.0170	16.11 ± 1.66
FNO + aug.	6	0.5009 ± 0.0590	0.5276 ± 0.0525	0.3816 ± 0.0322	18.76 ± 7.34
FNO + aug. + orbit	4	0.3676 ± 0.0360	0.3950 ± 0.0334	0.2677 ± 0.0174	16.87 ± 3.19

Table 2: Training compute accounting for the headline 2% N64 Galilean runs. Forward/backward counts and transformed-sample counts are computed from the logged optimizer steps; wall-clock minutes are estimated from immutable RNG checkpoint timestamps for these legacy cached runs, and new runs now write `train_wall_seconds` directly. Relative train time is normalized to the compute-matched supervised augmentation baseline.

Method	Opt. steps	Model fwds.	Bwd. passes	Aug. labeled samples	Orbit samples	Train min.	Rel. train time	Latency ms/sample
FNO	600	600	600	0	0	14.6 ± 0.4	$0.71\times$	16.53 ± 2.13
FNO + orbit	600	1200	600	0	9600	18.7 ± 0.6	$0.91\times$	16.11 ± 1.66
FNO + aug.	900	1800	900	14400	0	20.6 ± 2.5	$1.00\times$	18.76 ± 7.34
FNO + aug. + orbit	600	1800	600	9600	9600	21.0 ± 2.6	$1.02\times$	16.87 ± 3.19

Compute accounting. Table 2 reports training compute for the same headline runs. The compute-matched augmentation baseline uses 900 optimizer steps and 1800 estimated model forward passes, while augmentation plus orbit consistency uses 600 optimizer steps and the same 1800 estimated model forward passes because each step evaluates the original, augmented, and orbit-transformed inputs. Inference latency remains within run-to-run noise because the deployed map is still the same FNO2d backbone.

Label efficiency. Figure 1 and Table 3 show that the benefit persists across the low-label curve. Against compute-matched augmentation, $\lambda_{\text{orb}} = 0.10$ reduces orbit OOD error by 13.7%, 25.1%, and 22.3% at 1%, 2%, and 5% labels respectively. The corresponding equivariance-defect reductions are 21.3%, 29.8%, and 33.2%. This supports the label-efficiency framing: the orbit objective is most useful when supervision is scarce, while preserving the same inference map.

OOD severity. Figure 2 and Table 7 evaluate Galilean boosts from below the training radius to twice the training radius. The augmentation-plus-orbit model maintains a lower orbit OOD error at every tested boost. The relative OOD-error reductions are 26.4%, 25.8%, 25.3%, 23.6%, and 20.3% at maximum boosts 0.10, 0.20, 0.25, 0.35, and 0.50. Equivariance-defect reductions follow the same trend, from 33.6% at boost 0.10 to 23.6% at boost 0.50.

Canonicalization calibration. Table 4 reports a privileged relative-boost calibration, not a deployable analytic canonicalization baseline. In the OOD severity protocol, each test input a is paired with a sampled relative boost δ , producing $T_\delta a$. The calibration uses this sampled δ and the paired in-distribution prediction $\mathcal{G}_\theta(a)$, then reports $T_\delta \mathcal{G}_\theta(a)$. This is useful for measuring how much of the direct OOD error comes from motion along the tested orbit, but it assumes access to the paired source input and sampled relative transform. It should therefore not be confused with the observable-frame analytic baseline, which reads the absolute ambient velocity c from the input channels and canonicalizes a single test input. At maximum boost 0.50, augmentation plus orbit consistency has direct OOD error 0.4910 ± 0.0216 , improving substantially over augmentation 0.6158 ± 0.0384 while remaining above its privileged calibration value 0.3676 ± 0.0360 .

Defect as predictor. Figure 3 and Table 5 show that equivariance defect is not merely an auxiliary metric. Across 90 cached N64 Galilean observations, defect correlates strongly with orbit OOD relative L^2 (Pearson $r = 0.851$, Spearman $\rho = 0.838$); after residualizing source, method, label fraction, maximum boost, and λ_{orb} , the partial Pearson correlation remains $r = 0.779$. The relationship also holds within the OOD severity

Table 3: Label-efficiency comparison for compute-matched augmentation versus augmentation plus orbit consistency with $\lambda_{\text{orb}} = 0.10$.

	Labels	Method	Orbit OOD L^2	Eq. defect	Seeds
	1%	Aug.	0.7421 ± 0.0838	0.5080 ± 0.0456	3
	1%	Aug. + orbit	0.6406 ± 0.0508	0.3999 ± 0.0197	3
	2%	Aug.	0.5276 ± 0.0525	0.3816 ± 0.0322	5
	2%	Aug. + orbit	0.3950 ± 0.0334	0.2677 ± 0.0174	5
	5%	Aug.	0.3237 ± 0.0013	0.2427 ± 0.0014	3
	5%	Aug. + orbit	0.2515 ± 0.0112	0.1621 ± 0.0078	3

Table 4: Privileged relative-boost calibration at 2% labels. Direct OOD evaluates the model on the boosted input. The privileged calibration uses the sampled relative boost from the evaluation protocol and the paired in-distribution prediction, so it is not an observable-frame analytic baseline. Lower is better.

Max boost	Aug. direct OOD	Aug. priv. calib.	Aug. + orbit direct OOD	Aug. + orbit priv. calib.
0.10	0.5050 ± 0.0574	0.5009 ± 0.0590	0.3715 ± 0.0349	0.3676 ± 0.0360
0.20	0.5176 ± 0.0543	0.5009 ± 0.0590	0.3840 ± 0.0331	0.3676 ± 0.0360
0.25	0.5277 ± 0.0523	0.5009 ± 0.0590	0.3943 ± 0.0318	0.3676 ± 0.0360
0.35	0.5555 ± 0.0476	0.5009 ± 0.0590	0.4242 ± 0.0285	0.3676 ± 0.0360
0.50	0.6158 ± 0.0384	0.5009 ± 0.0590	0.4910 ± 0.0216	0.3676 ± 0.0360

curve alone ($r = 0.761$), supporting the mechanism that improved orbit consistency tracks improved OOD behavior.

Ablations and scope. Table 6 reports the orbit-weight ablation at 2% labels and supports the final $\lambda_{\text{orb}} = 0.10$ setting. Semi-supervised joint orbit training is useful as an appendix result because it reduces equivariance defect, but it is not the strongest predictive-error result. We do not include a 3D PDE in the main study: a same-resolution 3D analogue would multiply raw field size by 64 and introduce a new solver and validation burden. The current contribution is deliberately narrow: exact-symmetry validation plus a reproducible 2D Galilean low-label result.

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Table 5: Correlation between equivariance defect and orbit OOD relative L^2 across cached N64 Galilean seeds, label fractions, orbit weights, and boost severities. Partial Pearson r residualizes source, method, label fraction, maximum boost, and λ_{orb} .

Scope	Obs.	Pearson r	Spearman ρ	Partial r
All	90	0.851	0.838	0.779
Train/lambda/label	40	0.990	0.990	0.976
OOD severity	50	0.761	0.754	0.516

Table 6: Orbit-weight ablation at 2% labels. The final setting $\lambda_{\text{orb}} = 0.10$ gives the best completed N64 Galilean result.

λ_{orb}	Orbit OOD L^2	Eq. defect	Seeds
0.01	0.5768 ± 0.0378	0.3955 ± 0.0235	3
0.05	0.4518 ± 0.0397	0.3161 ± 0.0247	5
0.10	0.3950 ± 0.0334	0.2677 ± 0.0174	5

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Table 7: Relative reductions of augmentation plus orbit consistency over compute-matched augmentation across Galilean OOD severities at 2% labels.

Max boost	OOD reduction	Defect reduction
0.10	26.4%	33.6%
0.20	25.8%	31.3%
0.25	25.3%	30.2%
0.35	23.6%	27.8%
0.50	20.3%	23.6%

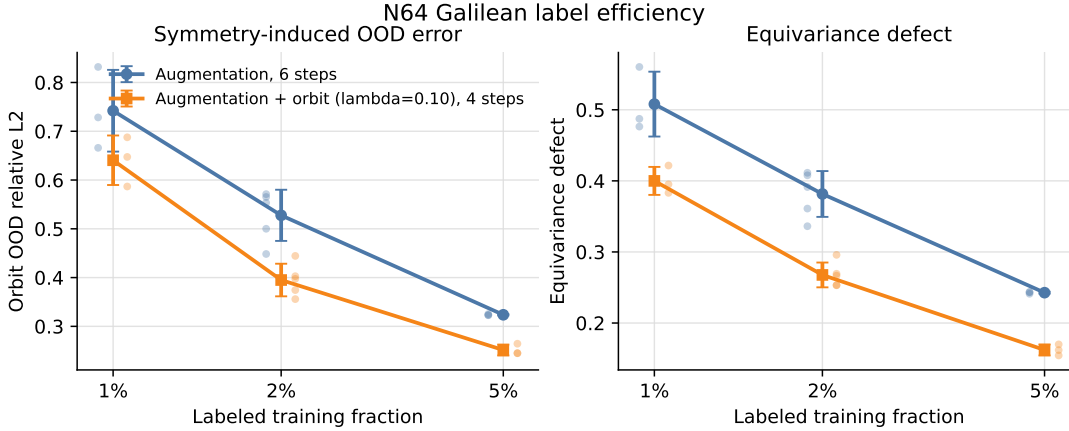


Figure 1: N64 Galilean label efficiency. Error bars show standard deviation across seeds and faint points show individual seeds. Augmentation plus orbit consistency improves orbit OOD error and equivariance defect at 1%, 2%, and 5% labels.

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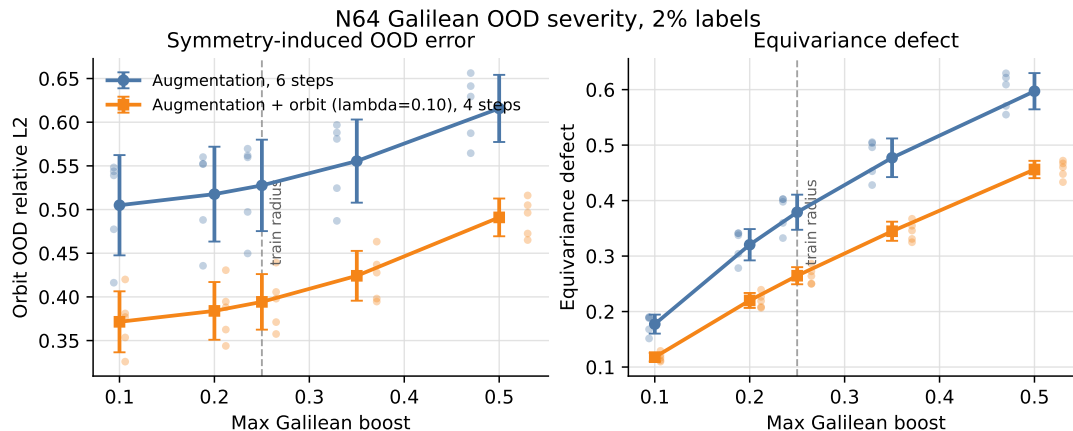


Figure 2: OOD severity at 2% labels. The dashed line marks the training boost radius. Augmentation plus orbit consistency remains better than compute-matched augmentation from below the training radius through twice the training radius.

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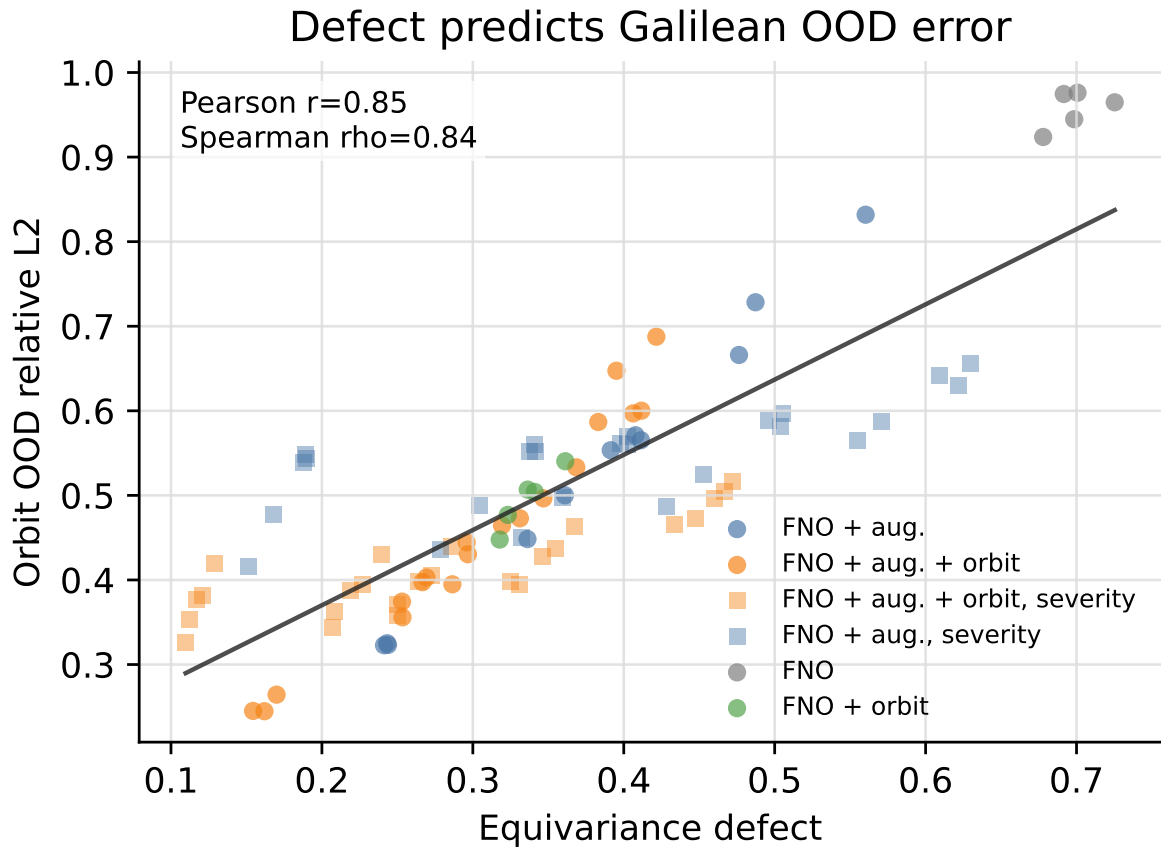


Figure 3: Equivariance defect predicts Galilean orbit OOD error across cached N64 seeds, label fractions, orbit weights, and boost severities. Circles are trained-model evaluations and squares are severity evaluations.